

Theory of Codes and Cryptography: Practice 1

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1. Provide the different features that the system should

meet. The system should meet the following features:

- **Confidentiality:** only the person with the key can access the information.
- **Identification or authentication of the entity:** making sure the user trying to decrypt the information is the right person.
- **Authorization:** Deciding whether you have the right to access the data and granting the right permissions.
- **Data integrity:** Checking if the data wasn't altered somewhere.
- **Signature:** It links a specific person to a given data ensuring that the person cannot later claim "I didn't send this".

2. Identify and characterize the different entities that interact in the system and describe their main features.

- **User:** person providing data to encrypt or person wanting to decrypt the data.
- **Administrator:** system manager that coordinates access.
- **Symmetric encryption algorithm:** the main algorithm responsible for encrypting and decrypting the data.
- **Public key encryption algorithm:** responsible for encrypting and decrypting the key which the main algorithm uses.
- **Authentication algorithm:** verifies the identity of the user and passes that information to the administrator.

3. Describe the provisional design of the project and explain how the information can be accessed only by the authorized users.

The provisional design is based on a hybrid cryptographic architecture.

- **Data Encryption:** The Symmetric Encryption Algorithm is used to efficiently encrypt the original Data, producing the Encrypted Data.
- **Key Encapsulation:** The Public Key Encryption Algorithm is used to securely encrypt the symmetric session key.
- **Identity Verification:** The Authentication Algorithm is utilized by the Administrator to verify the User's identity before processing requests.

The information can be accessed only by authorized users because the Administrator strictly controls the access flow through the Authentication Algorithm. Even if unauthorized entities intercept the Encrypted Data, they cannot read it without the symmetric key. This key is secured by the Public Key Encryption Algorithm and can be decrypted exclusively by the authorized User possessing the corresponding private key.

4. Indicate the information that should be additionally stored in order to properly access the original data.

To ensure that authorized users can correctly decrypt and recover the original content, the system must store specific cryptographic parameters and metadata alongside the encrypted file. The required information includes:

- Encrypted symmetric key: The symmetric key encrypted with the public key, allowing only the authorized user to decrypt it.
- Initialization vector (IV): A random value required by the symmetric algorithm to decrypt the data correctly.
- Authentication tag or signature: Data used to verify the integrity of the file and the user's identity.
- File metadata: Basic information such as original filename, file extension, and owner ID.

Storing this additional metadata guarantees that the system can execute the decryption process smoothly without compromising confidentiality or data integrity.

5. Programming language and libraries

The application will be developed in **Python 3**, using the following libraries:

- **Flask**: creates the web API and handles user requests.
- **cryptography**: provides AES encryption and decryption to protect file contents.
 - **liboqs-python**: provides ML-KEM (derived from Kyber) for establishing the shared encryption secret, and ML-DSA (derived from Dilithium) for digital signatures used to authenticate users.
- **secrets**: Python's built-in module for generating cryptographically secure random values, such as authentication challenges.